Unauthenticated vs Authenticated Scan

In this lab, we compare the results of unauthenticated and authenticated scans taken in Linux OS.

Setup

- **VM**: Linux (ubuntu 24.04)
- **Scanner**: Tenable Nessus

Things we do

We are going to perform an unauthenticated and an authenticated scan on our Linux virtual machine with an internal scan server, which are both located in the same virtual network. Compare the reports generated after the scannings.

Difference

- **Unauthenticated Scans**: Vulnerability Management tools do not possess your admin credential. Only perform surface-level evaluation. Mainly focus on external facing vulnerability findings, which is what usually attackers see.
- **Authenticated Scans**: Scanners perform in-depth vulnerability scans with a provisioned admin credential, going through the OS configuration, file system, etc.

Step

- **Start Scans**: Unauthenticated scan took 6 minutes, authenticated scan took 4 minutes.

Final Report

10.0.0.204

Summary

Critical	High	Medium	Low	Info	Total
0	0	0	1	22	23

Details

Severity	Plugin Id	Name
Low	10114	ICMP Timestamp Request Remote Date Disclosure
Info	277653	Remote Services Using Post-Quantum Ciphers
Info	11219	Nessus SYN scanner
Info	19506	Nessus Scan Information
Info	149334	SSH Password Authentication Accepted
Info	10267	SSH Server Type and Version Information
Info	25220	TCP/IP Timestamps Supported
Info	12053	Host Fully Qualified Domain Name (FQDN) Resolution
Info	181418	OpenSSH Detection
Info	54615	Device Type
Info	39520	Backported Security Patch Detection (SSH)
Info	110723	Target Credential Status by Authentication Protocol - No Credentials Provided
Info	10287	Traceroute Information
Info	153588	SSH SHA-1 HMAC Algorithms Enabled
Info	22964	Service Detection

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Summary

Critical	High	Medium	Low	Info	Total
1	2	5	2	59	69

Details

Severity	Plugin Id	Name
Critical	297660	Ubuntu 22.04 LTS / 24.04 LTS / 25.10 : Inetutils vulnerability (USN-7992-1)
High	299036	Ubuntu 22.04 LTS / 24.04 LTS / 25.10 : libpng vulnerability (USN-8039-1)
High	297802	Ubuntu 16.04 LTS / 18.04 LTS / 20.04 LTS / 22.04 LTS / 24.04 LTS / 25.10 : GNU C Library vulnerabilities (USN-8005-1)
Medium	298656	Vim < 9.1.2132 Buffer Overflow (GHSA-5w93-4g67-mm43)
Medium	298268	Ubuntu 22.04 LTS / 24.04 LTS / 25.10 : GLib vulnerabilities (USN-8017-1)
Medium	298266	Ubuntu 14.04 LTS / 16.04 LTS / 18.04 LTS / 20.04 LTS / 22.04 LTS / 24.04 LTS / 25.10 : Python vulnerabilities (USN-8018-1)
Medium	297658	Ubuntu 24.04 LTS : libpng vulnerabilities (USN-7993-1)
Medium	299331	Ubuntu 22.04 LTS / 24.04 LTS / 25.10 : GnuTLS vulnerabilities (USN-8043-1)
Low	10114	ICMP Timestamp Request Remote Date Disclosure
Low	299636	Ubuntu 22.04 LTS / 24.04 LTS / 25.10 : libssh vulnerabilities (USN-8051-1)
Info	181418	OpenSSH Detection
Info	149334	SSH Password Authentication Accepted
Info	45590	Common Platform Enumeration (CPE)
Info	70657	SSH Algorithms and Languages Supported